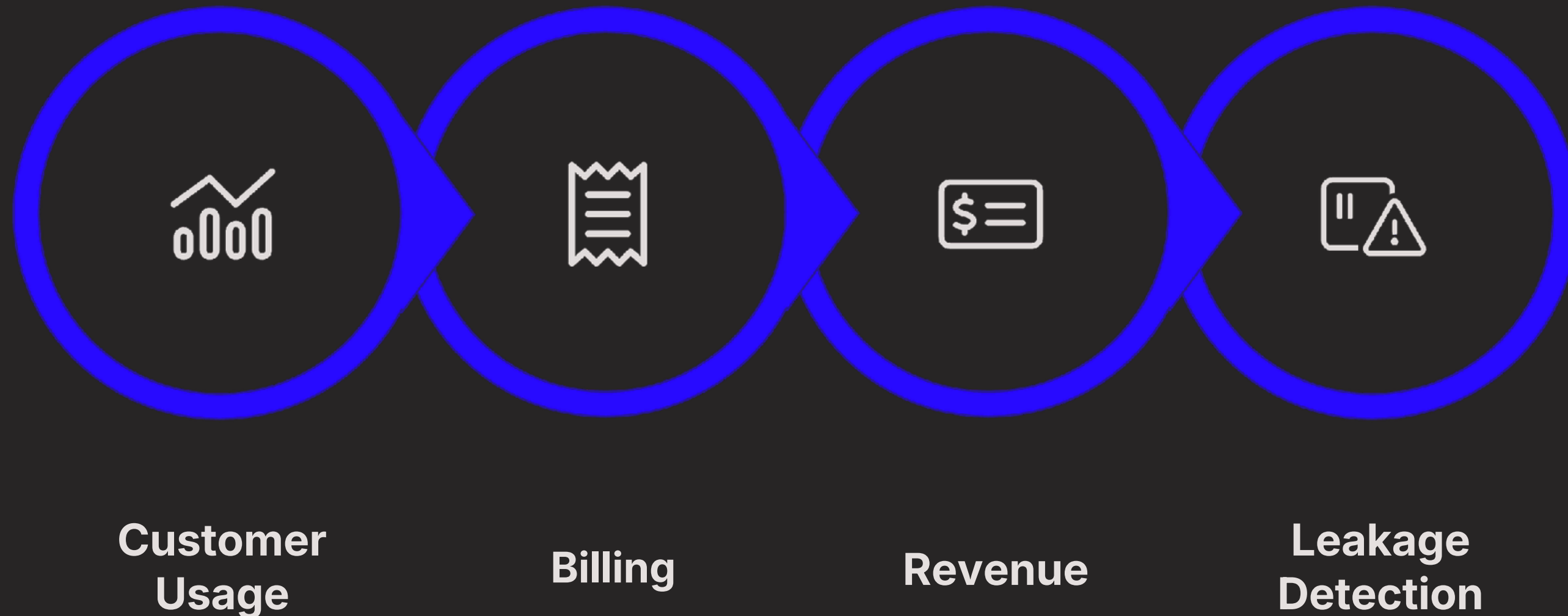


RevenueGuard: Telecom Revenue Leakage Detection

Using Usage and Billing Analytics to detect where expected telecom revenue is not being captured.



Why Revenue Leakage Matters

The Problem

Telecom companies handle thousands of customer transactions every month. Revenue is lost when:

- Usage is not billed
- Customers are under-billed
- Wrong tariffs are applied
- Plans do not match billing
- Roaming usage is missed
- Billing calculations contain anomalies

The Aim

Identify, quantify, and prioritize telecom revenue leakage by comparing expected revenue with actual billed revenue.

Objectives

Analyze Customer usage and billing data	Calculate Expected vs actual revenue
Detect Revenue leakage and leakage types	Assign Risk levels and priorities
Deliver Actionable Power BI insights	



From Telecom Usage to Leakage Detection

Dataset

1,000

Customers

12

Months

12,000

Records

Data Tables

- **Customers:** Customer, Region, Segment, Contract, Plan
- **Plans:** Monthly Fee, Data Allowance, SMS Allowance, Tariff Rates
- **Usage & Billing:** Data, Voice, SMS, Roaming, Billed Usage, Expected Revenue, Actual Revenue

Key Formulas

Revenue Leakage = Expected Revenue – Actual Revenue

Leakage % = (Revenue Leakage ÷ Expected Revenue) × 100



Where Is the Revenue Being Lost?

₹83.37L

Expected Revenue

₹69,441

Revenue Leakage

1,544

Leakage Cases

₹82.69L

Actual Revenue

0.83%

Leakage Rate

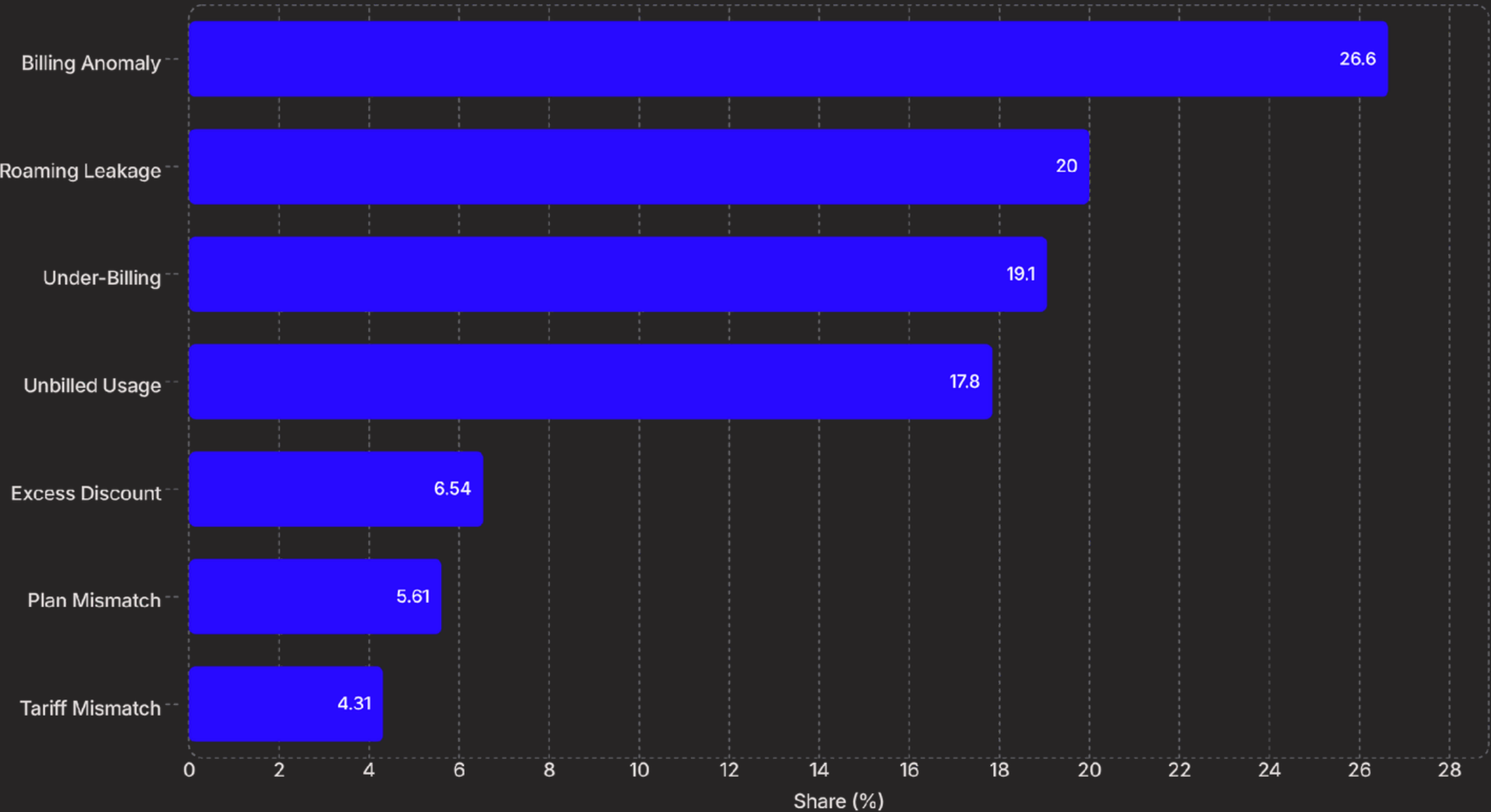
22

High-Risk Cases

Key Insight: Billing Anomaly, Roaming Leakage, Under-Billing, and Unbilled Usage together account for 83.54% of total financial leakage.

Leakage by Type

Leakage Type



RevenueGuard: Interactive Revenue Assurance Dashboard

Dashboard Capabilities



Revenue Comparison

Expected vs Actual Revenue with leakage tracking



Regional Analysis

Leakage by Region — South leads at Rs 19,999.64



Risk Prioritization

22 high-risk cases flagged for immediate investigation



Smart Filters

Filter by month, region, plan, and customer segment

Key Business Insights

26.64%

Billing Anomaly — largest single leakage contributor

Rs 19,999.64

South region — highest absolute regional leakage

83.54%

Usage and billing categories drive most financial leakage

22 Cases

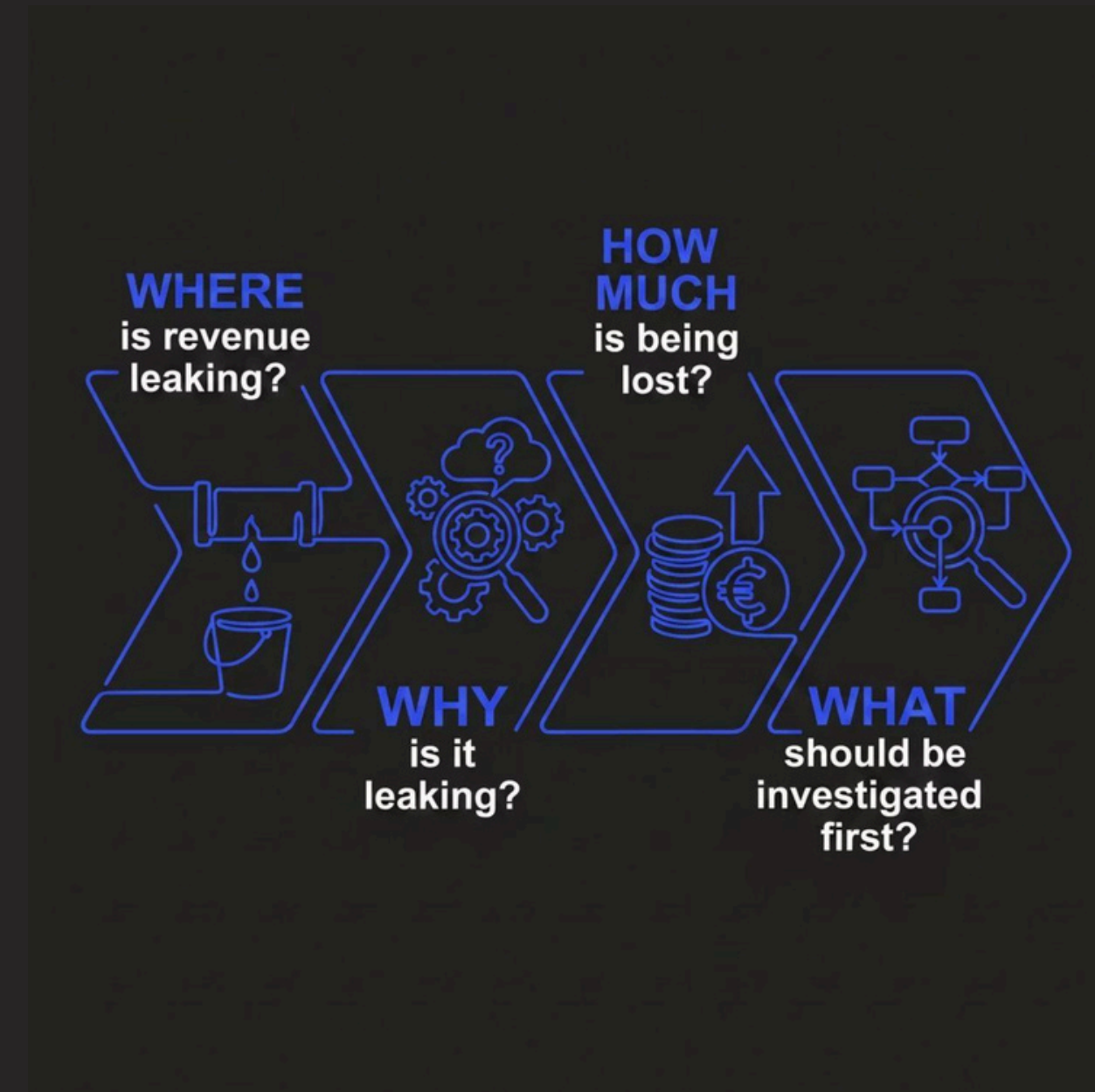
High-risk cases ready for immediate prioritization

From Detection to Revenue Recovery

Recommendations

- 1 Investigate high-value leakage first
- 2 Audit billing anomalies
- 3 Monitor unbilled usage
- 4 Strengthen roaming revenue assurance
- 5 Audit tariff and plan mapping
- 6 Continuously monitor leakage

The RevenueGuard Framework



Conclusion

RevenueGuard transforms telecom usage and billing data into actionable revenue-assurance insights.

Future Enhancement

Real-time monitoring combined with Machine Learning-based anomaly detection.